Installing Apache Web Server

Session 2



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Objectives

- *Identify the system requirements for installing Apache Web Server*
- *Explain the installation process of Apache Web Server*
- *Evaluate the installation of Apache Web Server*
- *Explain the process of editing the configuration file,* `httpd.conf`

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System Requirements for Installing Apache Web Server

- Hard Drive Space
  - Requires 50 MB free space on the hard drive
- Compiler
  - Requires the installation of ANSI-C compiler prior to Apache 2.2.17 installation
- Interpreter
  - Requires a Perl 5 interpreter for supporting scripts written in Perl such as, `apxs` or `dbmmanage`

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- Executed on a Linux platform with the root account privileges
- Consists of:
  - Unpacking the source code
  - Configuring
  - Compiling
  - Installing
- The main configuration file of Apache `httpd.conf` contains modules and directives

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Step 1

Login to Linux as root in the GNOME environment

Step 2

Download the `httpd-2.2.17.tar.gz` file from `http://httpd.apache.org/download.cgi`

Step 3

Right-click the `httpd-2.2.17.tar.gz` file and select `Extract Here`. The package contents are extracted to the folder named `httpd-2.2.17` in the current directory

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Step 4

Right-click the `httpd-2.2.17` folder and select `Open in Terminal`



Terminal Window

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Step 5

To configure Apache with the default options, enter the following command at the command prompt:

```
./configure --prefix=/usr/local/apache2
```



Output of configure Command

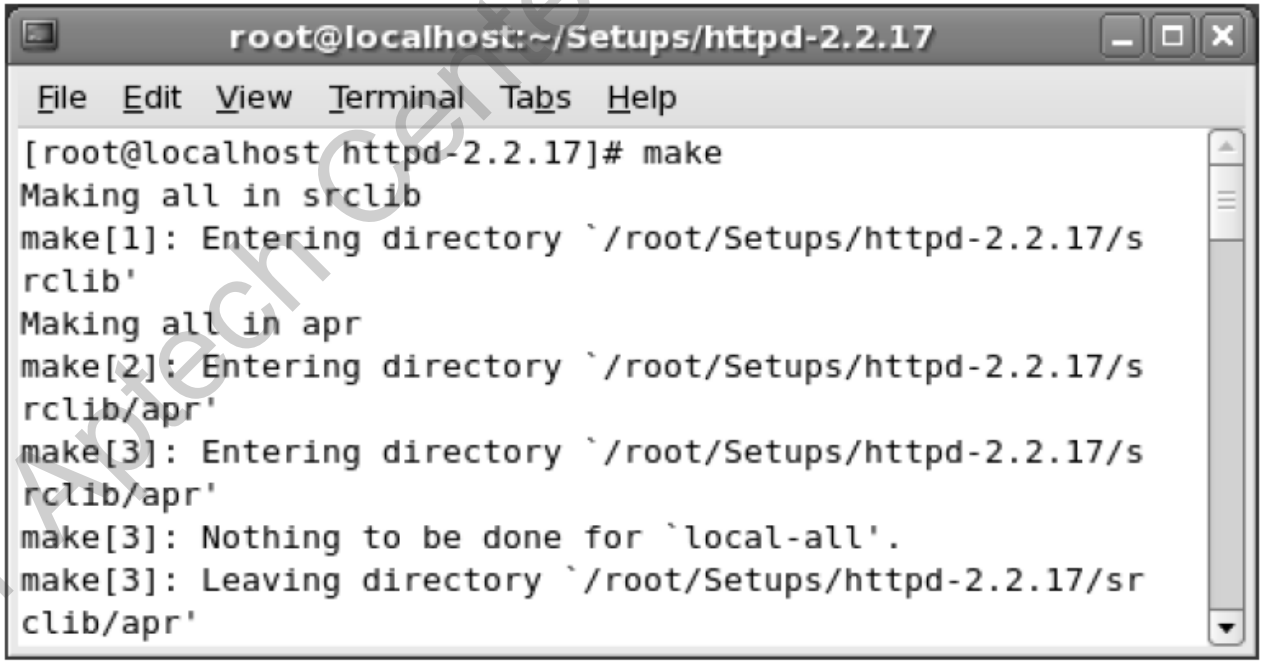
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Step 6

To compile the Apache source code, enter the following command at the command prompt

```
make
```



Output of make Command

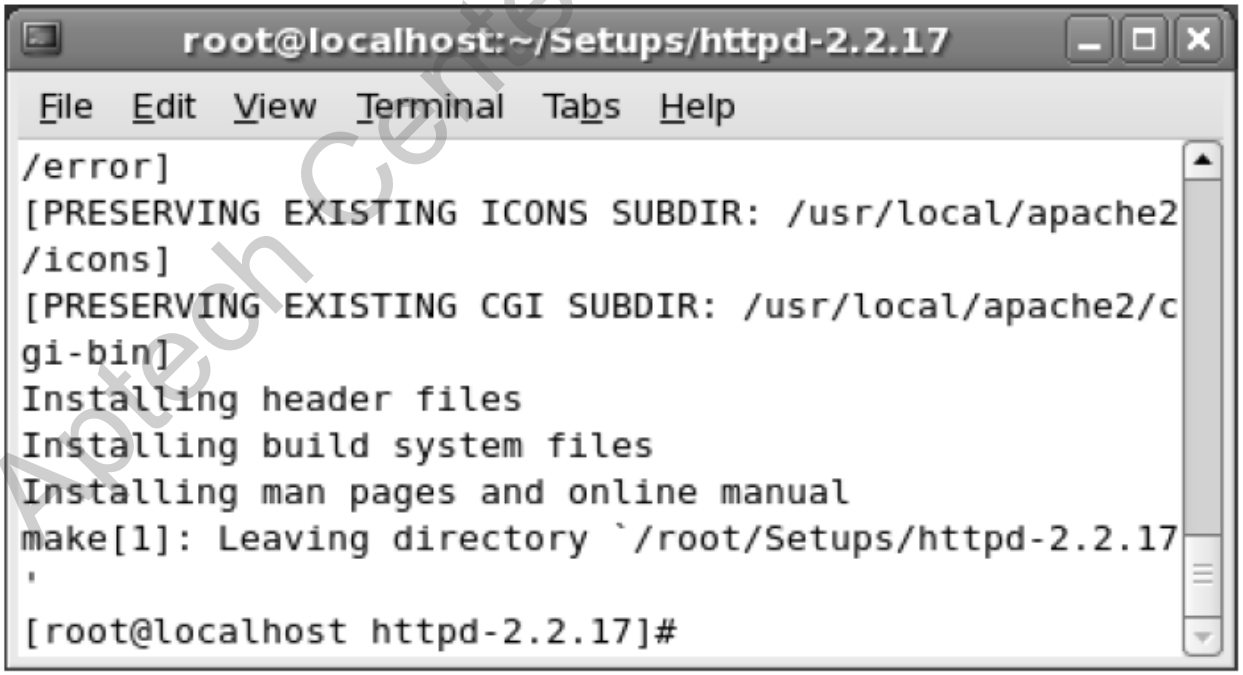
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Step 7

To install the resultant binary files and documentation to their respective directories, enter the following command at the command prompt:

```
make install
```



Output of make install Command

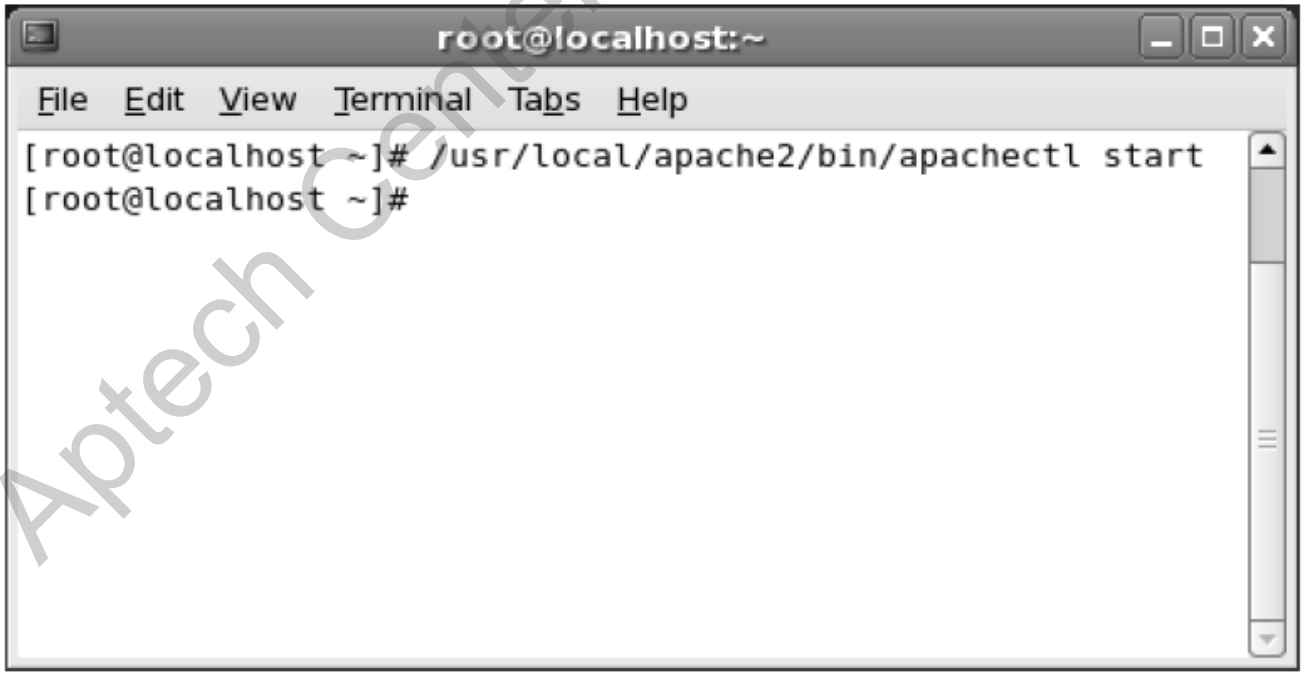
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Step 1

To start the Web server, enter the following command at the command prompt:

```
/usr/local/apache2/bin/apachectl start
```



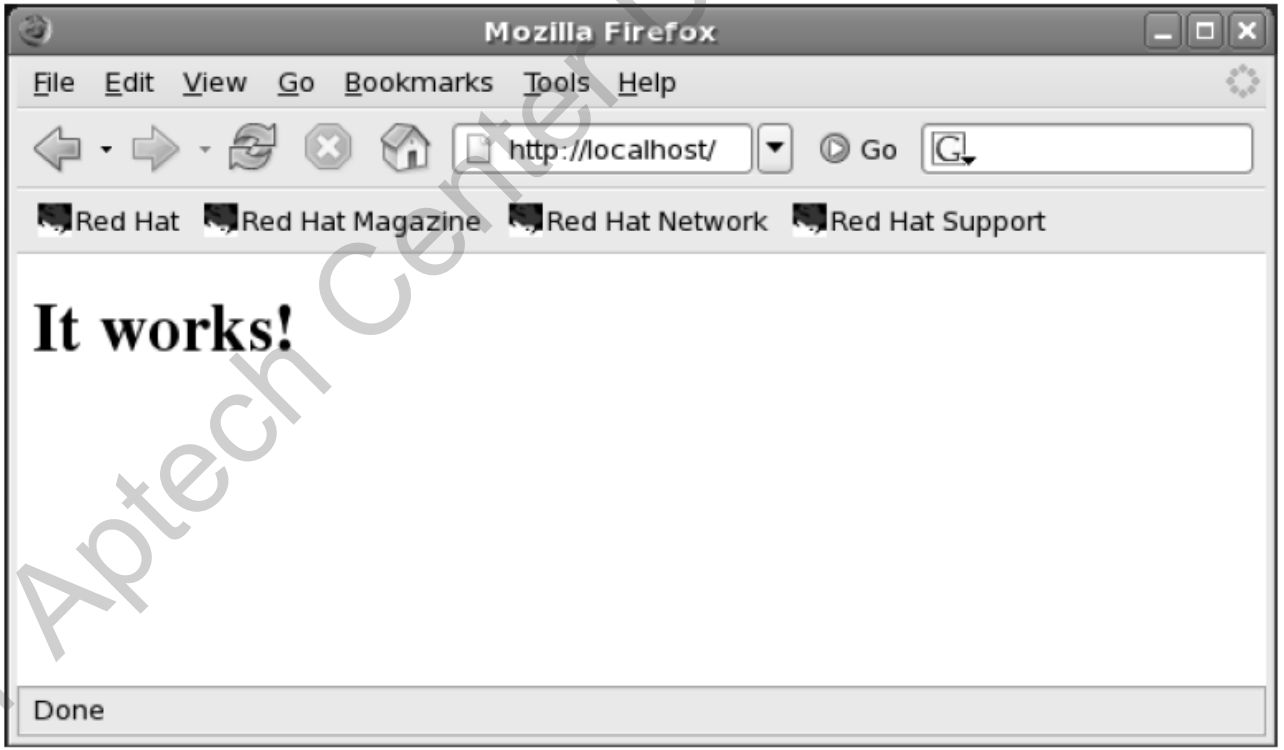
Starting Apache Server

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Step 2

Open the Mozilla Web browser and enter `http://localhost/` in the Address bar and press Enter



Local Host Page

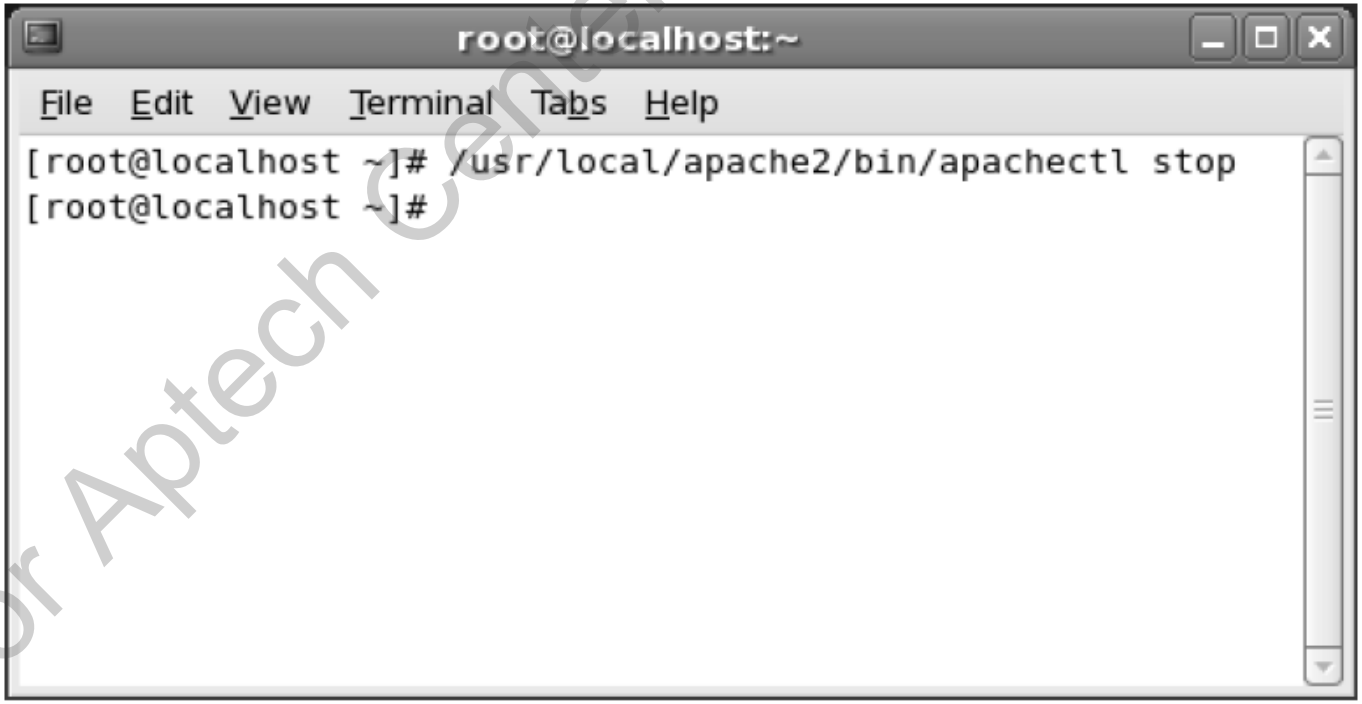
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Step 3

To stop the server, enter the following command at the command prompt:

```
/usr/local/apache2/bin/apachectl stop
```



Stopping Apache Web Server

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Editing httpd.conf File

- To open the `httpd.conf` file, perform the following steps:
  1. Login to Linux as root in the `GNOME` environment
  2. To start a new terminal, select Applications → Accessories → Terminal. The Terminal window is displayed
  3. To browse to the conf directory, enter the following at the command prompt:

```
cd /usr/local/apache2/conf
```

  4. To open the `httpd.conf` file in the vi editor, enter the following at the command prompt:

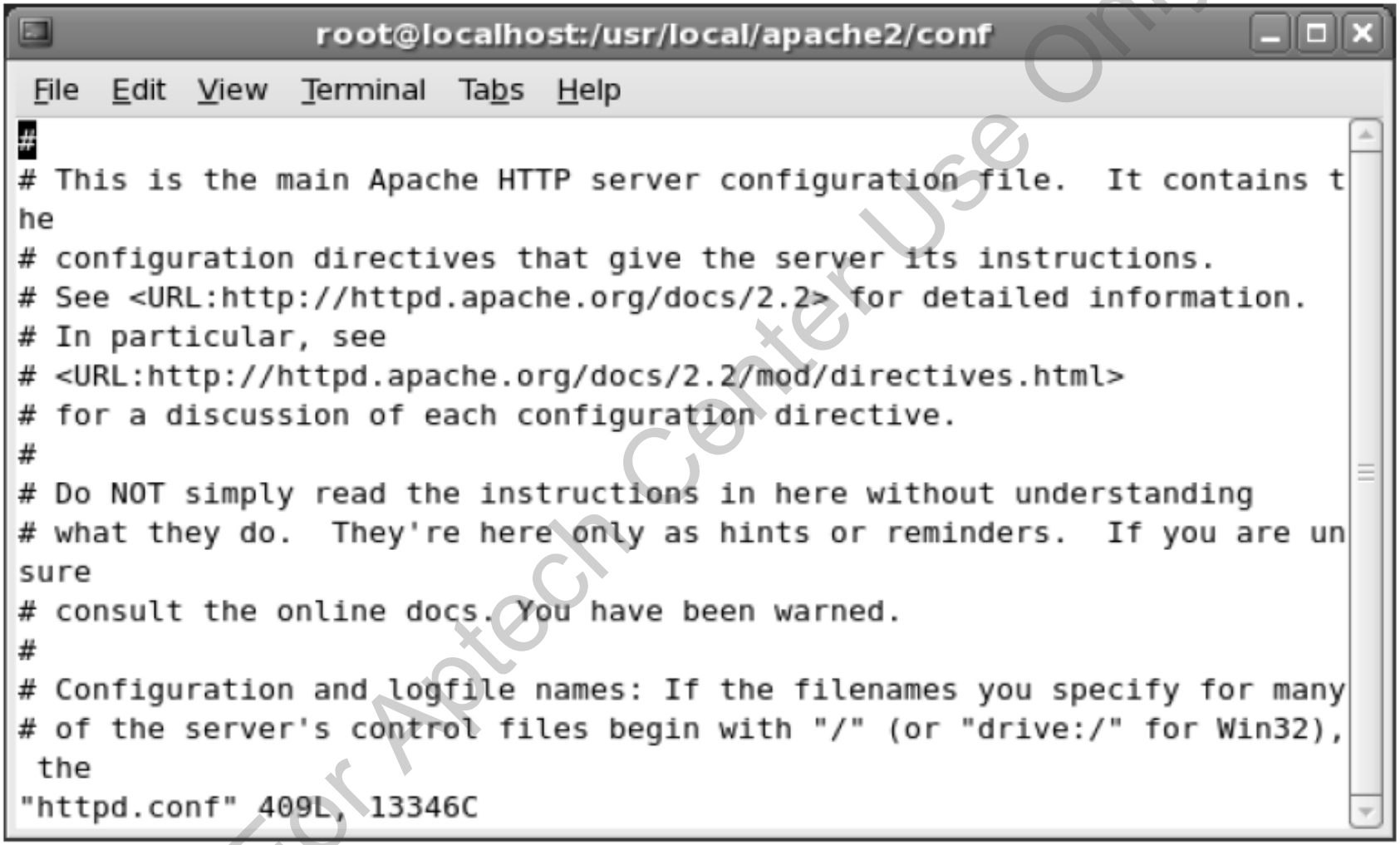
```
vi httpd.conf
```

  5. To save and exit the vi editor, enter the following at the command prompt:

```
:wq
```

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httpd.conf File

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Editing httpd.conf File

- The `httpd.conf` file consists of directives to configure the Apache server at startup
- The table shown describes some of the directives:

Directive Name	Description
`ServerName`	Specifies the hostname and the port for the server.
`Listen`	Instructs the server to listen to the specific IP addresses and ports.
`ServerRoot`	Specifies the directory path where the server is installed. The default option is `/usr/local/apache2`.
`DocumentRoot`	Specifies the directory path from where the server processes the files. The default option is `/usr/local/apache2/htdocs`.
`LanguagePriority`	Sets the preferred language.
`DefaultLanguage`	Defines the default language option.

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Summary

- Installing Apache includes:
  - Unpacking the source code
  - Configuring the Apache source tree
  - Compiling the source code to build executables
  - Installing Apache using the configure, make and make install command
- `httpd.conf` is Apache's main configuration file that contains modules and directives
- The `apachectl` control script invokes the `httpd` program

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